

Energy Consumption Project

You are a data scientist working for an energy analytics company. Your manager asked you to help better understand household electricity consumption in order to provide personalized advice to households and improve energy efficiency. You are given access to a dataset of 10,000 households with various features describing their living conditions, number of appliances, energy usage behavior, and more.

Your task now is to build models using different data mining techniques that can:

- Predict the daily energy consumption of new households.
- Segment households into meaningful groups based on their energy usage patterns

This analysis will help in offering targeted energy-saving recommendations and personalized marketing.

Different ways to go:

1. Use **visualization** techniques to analyze the features that most affect energy consumption (e.g., home size, appliance count, heating/cooling hours).
2. Use **clustering methods** to identify types of households with similar consumption patterns (e.g., “energy-efficient small homes”, “large inefficient homes”, etc.).
3. Use **supervised learning with feedforward neural networks** to predict the average daily energy consumption of a household.

In any case, leave out 10% of the data (1,000 households) to test your predictions and validate your insights. If your models show good performance, the company will integrate them into their smart energy platform for real-time recommendations.

You need to submit a **report** to your manager explaining:

- The dataset and feature descriptions
- The data mining methods you used (visualization, clustering, neural networks, etc.)
- The results and insights you obtained from each method
- Visualizations that support your analysis and findings

At the end, submit your **code** and the **report** to your manager as a complete analysis package.